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COLLEGE OF ENGINEERING
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**UNIVERSITY OF BATANGAS – LIPA CAMPUS
COLLEGE OF ENGINEERING AND ARCHITECTURE
COMPUTER ENGINEERING**

24-Second Shot Clock

Final Project
Logic and Circuit Design

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PROFESSOR



Objectives:

At the end of this project, you should be able to:

1. To be able to design and construct a functional 24-second shot clock using digital logic components.
2. To be able to implement a countdown timing system using 74LS192 up/down counters and a 555 timer.
3. To be able to apply logic gates to limit the counting range between 24 seconds and 00 seconds.
4. To be able to interface BCD counters with seven-segment displays using the 74LS47 decoder.
5. To be able to integrate control features such as reset, pause, and buzzer activation into the shot clock system.

Equipment and Materials:

The following equipment and materials are needed in performing this laboratory exercise:

- 1 - DC Power Supply with +5V output capability
- 1 - Digital Multimeter
- 3 – Breadboard
- 2 – 74LS192 for count-up and down
- 2 – 74LS47 BCD Decoder
- 1 - 7404 or 74LS04 hex Inverter IC
- 2 - 7408 or 74LS08 Quad 2-input AND gate IC
- 2 - 7432 or 74LS32 Quad 2-input OR gate IC
- 1 - 7400 or 74LS00 Quad 2-input NAND gate IC
- 1 - 7402 or 74LS02 Quad 2-input NOR gate IC
- 2 – 7 Segment Display (Common Anode)
- 2 – Slide Switch
- 2 – Momentary Push Buttons



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2 – DPST Switch

Presensitized PCBs

Ferric Chloride

Resistors

Capacitors

Connecting Wires

Discussion:

A 24-second shot clock is a digital timing system used in basketball to regulate the duration of an offensive possession. The system is designed to count down from 24 seconds to 00 while providing clear visual feedback through numerical displays and an audible alert when time expires. Its design highlights the practical application of digital logic, timing circuits, and control mechanisms in a real-time electronic system.

The countdown operation is achieved using 74LS192 up/down counters, which allow controlled counting in both directions and are configured specifically for downward counting. A 555 timer generates a stable one-second clock pulse that drives the counters. Logic gates are used to limit the count range between 24 and 00, preventing invalid states and ensuring proper rollover control. A pause button is integrated into the clock signal path to temporarily stop the countdown without resetting the current count, while a reset button initializes the counters back to 24 seconds to begin a new timing cycle.

For visual output, 74LS47 BCD-to-seven-segment decoder drivers convert the counter outputs into signals suitable for seven-segment displays, allowing the remaining time to be clearly shown. When the count reaches 00, combinational logic detects the zero condition and disables further counting while simultaneously activating a buzzer to indicate shot clock expiration. This integrated design demonstrates how counters, timers, decoders, Computer Engineering - CP314E



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and logic gates work together to form a functional and reliable 24-second shot clock system.

Procedure:

1. The project begins by creating the complete circuit schematic in Multisim, where a NE555 is configured as an astable clock to produce a one-second timing pulse.
2. Two 74LS192 ICs are included in the design to handle the units and tens digits, both arranged to operate in countdown mode using the clock signal from the 555 timer.
3. The BCD outputs of each counter are connected to a 74LS47 so that the numerical values can be displayed on common-anode 7-segment displays.
4. Logic is incorporated to recognize the BCD combination corresponding to 24, allowing the counters to be preset or loaded to this value whenever the reset control is activated.
5. Additional logic is added to detect the 00 condition, which inhibits the clock signal and prevents the counters from continuing below zero.
6. The full circuit is then simulated in Multisim to observe the countdown behavior and verify that the display transitions correctly from 24 to 00 at one-second intervals.
7. After successful simulation, the same circuit is assembled on a breadboard using actual components. Then checked for errors and debugging
8. The verified schematic is finally recreated in EAGLE CAD, where the layout is completed and prepared for etching.



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Results and Discussion:

The implementation of the 24-second shot clock was successful at the simulation stage, where the circuit consistently counted down from 24 to 00 with correct one-second timing. The logic designed to restrict the count within the valid range functioned properly, ensuring that the display never showed invalid values and that the countdown stopped at zero as intended. Simulation results confirmed that the integration of counters, logic gates, and display drivers worked cohesively, validating the theoretical design and demonstrating a clear understanding of digital timing and control concepts.

However, translating the simulated design into actual hardware revealed several practical challenges that affected full verification. The large number of interconnected components increased wiring complexity on the breadboard, making the circuit prone to loose connections, miswiring, and unstable signals. These issues required repeated troubleshooting and extended debugging sessions, often conducted overnight with classmates under significant time pressure, limited rest, and ongoing academic demands such as examinations. And making the circuit board using presensitized PCB, also does not turn up well. Although not all hardware issues were completely resolved, the group was able to confirm the fundamental operation of the shot clock and gained valuable experience in handling real-world limitations, highlighting the gap between ideal simulation environments and actual electronic implementation, as well as the importance of persistence and teamwork in engineering projects.



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Conclusion:

This project successfully demonstrated the design of a functional 24-second shot clock using digital logic components, with correct operation verified through simulation. The countdown timing system using a 555 timer and 74LS192 up/down counters effectively achieved a stable one-second countdown from 24 to 00. Logic gates were properly applied to limit the counting range and prevent invalid display values. The interfacing of BCD counters with seven-segment displays using the 74LS47 decoder provided clear and accurate visual output. Control features such as reset, pause, and buzzer activation were integrated into the system to enhance usability and functionality. Despite the circuit working reliably in simulation, the actual hardware implementation revealed practical challenges such as wiring complexity and signal instability. These difficulties emphasized the importance of careful construction, systematic debugging, and real-world testing. Overall, the project strengthened the group's technical skills, problem-solving ability, and teamwork under demanding time and academic constraints.



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Documentation:

